

Indirect Heat Integration across Plants: Novel Representation of Intermediate Fluid Circles

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Supporting Information

ABSTRACT: Interplant heat integration (IPHI) using intermediate fluid circles provides an additional opportunity for energy saving and emission reduction, beyond traditional intraplant heat integration. However, the number of intermediate fluid circles as well as their heat capacity and temperatures are all variables. It is difficult to find promising solutions, since such conditions generally lead to nonlinearity. To deal with this problem, a transshipment type model, which holistically optimizes the parameters and interconnectivity patterns of intermediate fluid circles and keeps the constraints linear, is developed. New formulations are developed to identify exact distributions of intermediate fluid circles (namely heat transfer available intervals) among all temperature intervals, with unknown temperatures. Additionally, mixed-integer linear constraints are developed to describe interconnectivity patterns of intermediate fluid circles so as to calculate piping and pumping cost. The performances of different interconnectivity patterns are explored, while not only the energy efficiency but also the total cost are optimized. Case studies show that better results with lower TAC and appropriate interconnectivity patterns of intermediate fluid circles can be obtained, compared to the literature.

INTRODUCTION

The rising energy demand, the increasing price of fossil fuels, and the growing environmental concern highlight the significance and urgency of improving industrial energy efficiency. Other than deep understanding of fossil fuels,¹ heat integration (HI) is a well-established and effective technique for energy conservation. Heat integration can improve energy efficiency by utilizing waste heat, instead of directly optimizing fuel gas systems.² In the past decades, researchers and engineers have applied heat integration to design integrated energy systems ranging from processes within one plant to industrial clusters, considering sustainability,³ cost, and environmental impacts.^{4,5} The collaboration among industrial clusters, such as ecoindustrial parks (EIP), can further reduce the energy consumption and other resources. There are successful case studies of existing complexes in Yeosu, South Korea⁶ and Kashima Industrial Park in Japan,⁷ which show that the amount of energy consumption can be reduced significantly. Heat integration across multiple plants can be referred to as total site heat integration (TSHI) or interplant heat integration (IPHI). Typically, TSHI is defined as heat integration through a central utility system, while IPHI focuses on interplant heat exchange opportunities carried out either directly using process streams across plants or indirectly using intermediate fluid circles.

Compared to direct heat transfer, smaller heat saving can be achieved by indirect heat transfer, owing to multiple minimum temperature approaches. However, an intermediate fluid with a larger heat capacity than process streams will result in less piping and pumping cost. More importantly, considering the controllability and safety of pumping process streams over a long distance, indirect heat transfer would be a better choice.

Pinch analysis (PA) and mathematical programming (MP) are the two systematic methods for the heat integration. PA was first introduced by Linnhoff and Hindmarch,⁸ and it provided insight of potential energy saving opportunities in an individual plant. Ahmad and Hui⁹ extended PA to find maximum energy recovery between areas of integrity by the overlapping of grand composite curves. Both indirect heat integration using different levels of steam and direct heat integration using process streams were studied. In their subsequent research,¹⁰ the overall cost trade-off among energy, heat exchange area, and the number of interconnections was considered. The first work in TSHI was proposed by Dhole and Linhoff¹¹ to determine additional energy reductions across

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multiple plants at one site using site source sink profiles (SSSP) based on PA. Varbanov et al.¹² extended TSHI to include renewable energy sources, such as solar, wind, and biomass, accounting for the variable supply and demand. They also modified the total site targeting procedure¹³ to allow a temperature difference ($\Delta T_{\min}$) to be specified for each process on the site, obtaining more realistic heat recovery targets for the total sites. Then, Chew et al.^{14,15} extended the scope of PA for process modification of TSHI to maximize energy saving¹⁴ and reduce capital cost.¹⁵ Recently, Tarighaleslami et al.¹⁶ developed a new TSHI targeting methodology addressing non-isothermal utilities targeting incorporated isothermal utilities targeting in the same procedure, using industrial cases. Song et al.¹⁷ presented a novel screening algorithm, named NLQSA, for IPHI, based on PA and the theoretical maximum interplant heat recovery potential. Large-scale problems could be divided into several smaller sections using NLQSA, followed by a modified MINLP model to minimize the total annual cost (TAC) of each section. Also, they¹⁸ proposed an improved targeting procedure to determine the maximum interplant heat recovery potential for the indirect IPHI with the parallel connection pattern among three plants. Hong et al.¹⁹ developed a new graphical tool, namely, heat transfer block diagrams, where the utility consumption can be targeted, and cost-effective heat exchanger networks (HENs) can be constructed by analyzing heat surpluses and deficits on the diagram. They²⁰ also proposed a transformation method for HENs with different structures based on T–H and H–F diagrams.²¹ Although PA-based methods are proficient in targeting utility consumption and providing physical insights, the implementation of PA-based methods relies on heuristics and experiences and, more importantly, cannot provide the optimal design.

Mathematical programming methods can optimize single or multiple objective functions, such as utility consumption, heat exchanger number, and total annual cost. The three basic models for the synthesis of HENs are the transshipment model of Papoulias and Grossmann,²² the superstructure representation of Floudas et al.,²³ and the stage-wise superstructure of Yee and Grossmann.²⁴ On the basis of the transshipment model, Rodra and Bagajewicz²⁵ proposed a systematic procedure, based on linear programming (LP) and mixed-integer linear programming (MILP) models, to target maximum energy saving and determine the optimum location of intermediate fluid circles, respectively, for the system of two plants using both direct and indirect heat integration. Subsequently, this procedure was extended for the system of a set of plants.²⁶ Bada and Bandyopadhyay²⁷ addressed the minimization of the flow rate of thermal hot oil for indirect heat integration by an LP model. However, the establishment of interplant heat integration entails the high piping cost and pumping cost caused by long distance transportation. They should be holistically considered and optimized in the model. Chang et al.²⁸ developed a single stage superstructure of indirect interplant heat integration between source and sink plants using a hot water circuit as an intermediate fluid medium, considering piping cost, pumping cost, and heat loss. Subsequently, Chang et al.^{29,30} developed two mixed-integer nonlinear programming (MINLP) models for interplant direct heat integration²⁹ and indirect heat integration,³⁰ respectively, based on the stage-wise superstructure.²⁴ Most of factors, such as energy cost, heat exchanger cost, piping cost, and pumping cost, were simultaneously considered in these two models.

Besides the minimization of the utility consumption or the investment cost, the practical challenges of IPHI should be addressed, such as the sharing the benefits, heat exchanger locations, and so on. Nair et al.³¹ proposed a practical and rational strategy for direct IPHI in an ecoindustrial park (EIP), ensuring an identical rate of return on investment for all participating plants. In their subsequent research,³² the location of heat exchangers was optimized by an MINLP model, where heat exchangers can be located at any plant or central sites. Tan et al.³³ proposed an approach to determine the optimal allocation of costs and benefits in cooperative interplant process integration in EIP by an LP model. Cheng et al.³⁴ developed a game-theory-based optimization strategy to configure the optimal indirect IPHI schemes, including four consecutive steps to determine the utility cost, the heat flows among plants, the number of heat exchangers, and the network configuration. Then, this game-theory-based strategy was extended to consider only indirect IPHI by Chang et al.³⁵ Recently, Jin et al.³⁶ developed a superstructure-model-based two-stage procedures for direct IPHI problems, addressing the benefit allocation issues according to risk-based Shapley values. However, only the utility cost and the investment cost of heat exchangers were considered without the piping and pumping cost, which is significant in IPHI problems. Pan et al.³⁷ developed a multiplant indirect heat integration model based on a stage-wise superstructure²⁴ and proposed an Alopex-based evolutionary algorithm. Their model is efficient in solving small-scale problems but not large-scale problems.

In the present research about indirect heat transfer by intermediate fluid circles, the transshipment model²² and the stage-wise superstructure²⁴ were adopted. Although the stage-wise superstructure²⁴ was applied to minimize the total annual cost, the isothermal mixing assumption on each stage was generally adopted to make the model easier to solve, at the cost of missing better HENs with the non-isothermal mixing. In addition, several alternative structures are neglected in the stage-wise-based models. On the other hand, the traditional transshipment model²² can include non-isothermal mixing, and it can be formulated as an LP or MILP model to target energy consumption and find the heat transfer matches. Recently, our transshipment type model^{38,39} was proposed to minimize the total annual cost and obtain the network structures, with a newly developed mass flow pattern and heat flow pattern. Nonetheless, the model only addressed the synthesis of intraplant HEN.

From the intraplant to the interplant case, conditions of the intermediate fluid circle should be addressed. For example, the parameters and interconnectivity patterns of intermediate fluid circles will directly influence the performance of heat integration, such as the energy efficiency and total annual cost, which should be optimized in the model. Compared to the process streams, neither the starting nor the ending temperature of intermediate fluid circles is known. Modeling with such conditions by stage-wise superstructure²⁴ generally led to nonlinear terms. When using the transshipment model, energy balance formulations can be linear. But it will be challenging to identify the exact temperature interval distribution of intermediate fluid circles with unknown temperatures. Moreover, the interconnectivity pattern of intermediate fluid circles should be described in the model for the calculation of pumping and piping cost. To handle these two challenges, a new transshipment type HEN model is developed for the synthesis of intra and interplant HEN. New

mixed-integer linear formulations are developed to automatically identify the exact temperature interval distribution for intermediate fluid circles. In the proposed model, intermediate fluid circles can exchange heat with process streams from different plants in each available temperature interval. Novel mixed-integer linear formulations are developed to follow up on the interconnectivity patterns of intermediate fluid circles according to the heat transfer match in each temperature interval so that different heat integration possibilities can be explored, and the interconnectivity patterns can be optimized. The whole model is formulated as an MINLP model. All constraints in the model keep linear, while the objective function is nonlinear. Not only the energy consumption but also the heat exchanger cost, piping cost, and pumping cost are holistically and simultaneously optimized. One of the most significant features is that non-isothermal mixing is included while keeping all constraints linear. It makes the model easier to solve. Three literature examples are adopted to illustrate the applicability and effectiveness of the proposed model. Better results with lower TAC can be obtained, whereas multiple utilities are adopted in the industrial example to reduce the utility cost.

PROBLEM STATEMENT

The general problem for the synthesis of intra and interplant heat exchanger networks (IHENS) can be stated as follows. Given are several individual plants (P) segregated in different locations. In each plant, a set of hot process streams (HP), a set of cold process streams (CP), a set of hot utilities (HU), and a set of cold utilities (CU) are given, as well as their corresponding parameters. The objective is to design the intra and interplant HENs using intermediate fluid circles by minimizing the total annual cost, including the operating cost, the capital cost of heat exchangers, the piping cost, and the pumping cost. The topology structure of both the intra and interplant HEN, the flow rate of streams and the heat load of each heat exchanger, and the parameters of intermediate fluid circles will be determined.

Let l represent hot streams (process streams and utilities), m represent cold streams (process streams and utilities), li represent hot intermediate fluid streams, and mi represent cold intermediate fluid streams. Note that when li is equal to mi , it means that the hot and cold intermediate fluid streams belong to the same circle. For ease of representation, we define the following sets: H_p = (hot streams in plant p), HP_p = (hot process streams in plant p), HU_p = (hot utility streams in plant p), $H_p = HP_p \cup HU_p$, C_p = (cold streams in plant p), CP_p = (cold process streams in plant p), CU_p = (cold utility streams in plant p), $C_p = CP_p \cup CU_p$, $HI = (li|li \text{ is a hot intermediate fluid stream})$, $CI = (mi|mi \text{ is a cold intermediate fluid stream})$.

REPRESENTATION OF INTERPLANT HEN AND MATHEMATICAL MODEL

In this research, our transshipment type model,³⁸ which addresses the synthesis of intraplant HEN, is modified to include interplant heat integration using intermediate fluid circles. An intermediate fluid circle, which is divided into two substreams, is shown in Figure 1, where the intraplant HENs are simplified. Mass flows can exist between hot/cold substreams in each temperature level, and each hot/cold intermediate fluid substream can exchange heat with the cold/hot hot process stream of each plant, which results in a high degree

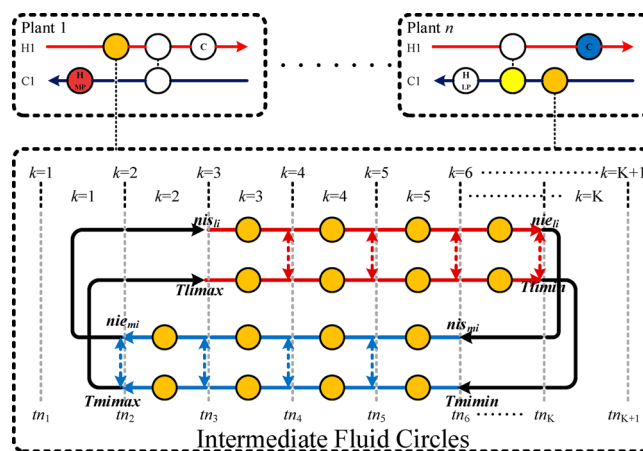


Figure 1. Representation of interplant HEN integration using intermediate fluid circles.

of freedom interconnectivity pattern of intermediate fluid circles. The interconnectivity pattern of intermediate fluid circles will determine the performance of interplant heat integration. Additionally, the interconnectivity pattern has a significant impact on the piping and pumping costs caused by long distance transportation between plants. In the proposed model, the interconnectivity pattern of intermediate fluid circles will be widely explored. The main work of this research is to develop new formulations for intermediate fluid circles based on the transshipment model while keeping constraints linear. In the following sections, the mathematical model will be briefly introduced, while the formulations for intraplant HEN can be found in the [Supporting Information](#).

Temperature Interval Distribution of Intermediate Fluid Circles. In the transshipment type HEN models, the first step is to construct the temperature intervals. In this model, temperature intervals ($k \in TI$) were constructed according to the starting temperature of process streams and utilities and ΔT_{min} . It is assumed that the temperatures of all hot streams are subtracted by ΔT_{min} . K temperature intervals will lead to $K + 1$ temperature levels, as shown in Figure 1. Then, the temperature interval distribution (briefly named interval distribution) of each process stream and utility stream among the whole temperature intervals can be easily determined, since their starting and ending temperatures are constant. However, unlike process streams, the starting and ending temperatures and the flow rates of hot and cold intermediate fluid streams are both unknown a priori, which means that the interval distribution of intermediate fluid circles is not fixed. Despite that, the temperature bounds of both hot and cold intermediate fluid streams can be determined. As for hot intermediate fluid streams, the minimum ending temperature T_{limin} can be determined by the minimum starting temperature of all cold process streams ($T_{limin} = \min(m, tcs_m)$). Since the intermediate fluid streams are transferred as circles, the minimum starting temperature of cold intermediate fluid streams T_{mimin} can be determined by T_{limin} . Accordingly, T_{mimin} should be equal to $T_{limin} + \Delta T_{min}$, considering that the temperatures of hot streams are subtracted by ΔT_{min} . In addition, the maximum starting temperature of hot intermediate fluid streams T_{limax} can be determined by the maximum ending temperature T_{mimax} of cold intermediate fluid streams ($T_{limax} = T_{mimax} - \Delta T_{min}$), while T_{mimax} can be determined according to the maximum ending

temperature of all hot process streams ($T_{mimax} = \max(l, the_i)$). Then, the starting temperature level (nis) and the ending temperature level (nie) of hot and cold intermediate fluid streams can be determined according to T_{limin} , T_{limax} , T_{mimin} , and T_{mimax} , as shown in Figure 1. Note that although the intermediate fluid circles may exist in all of these temperature intervals, these temperature intervals are not the exact temperature interval distribution for intermediate fluid circles. The reason is that hot/cold intermediate fluid streams may not be cooled down/heated up to T_{limin}/T_{mimax} . It means that they may only exist in some of the temperatures intervals, which are available for heat transfer. Thus, new formulations should be developed to identify heat transfer available temperature intervals for intermediate fluid circles automatically. It will be introduced in the section of heat flow pattern. Note that, in the following section, we will take hot intermediate fluid streams as an example to introduce mass flow pattern, heat flow pattern, and the interconnectivity pattern of intermediate fluid streams.

Mass Flow Pattern of Intermediate Fluid Circles. With the interval distribution of intermediate fluid streams among the temperature intervals, the mass flow pattern of intermediate fluid streams can be described. The flow rate sum of substreams belonging to hot intermediate fluid stream li is equal to the flow rate of hot intermediate fluid stream li

$$\sum_{ls' \in LS} flisk_{li,ls,k} = flif_{li} \quad \forall li \in HI, k \in TI, nis_{li} \leq k < nie_{li} \quad (1)$$

Also, the mass balance between adjacent temperature intervals for hot substreams is given considering the importing mass flows and exporting mass flows, as shown in Figure 2-(a)

$$flisk_{li,ls,k} = flisk_{li,ls,k-1} + \sum_{ls' \in LS} flir_{li,ls',k} - \sum_{ls' \in LS} flir_{li,ls',k} \quad \forall (li, ls) \in HILS, k \in TI, nis_{li} < k < nie_{li} \quad (2)$$

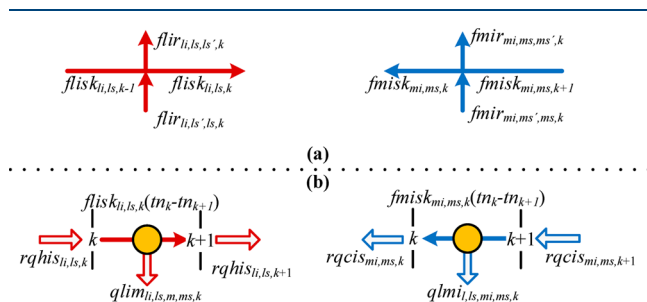


Figure 2. Mass flow pattern and heat flow pattern of intermediate fluid streams.

Since intermediate fluid streams are transferred in circles, the mass balance formulation between hot and cold intermediate fluid substreams is given by eq 3. As shown in Figure 1, the flow rate of hot intermediate fluid substream li, ls in the starting temperature interval is equal to the flow rate of corresponding cold intermediate fluid substream mi, ms in the ending temperature interval and the importing mass flows, subtracting the exporting mass flows.

$$flisk_{li,ls,k} = fmisk_{mi,ms,k} + \sum_{ms' \in LS} fmir_{mi,ms',k} - \sum_{ls' \in LS} fmir_{mi,ms',k} \quad \forall (li, ls) \in HILS, (mi, ms) \in CIMS, k, k' \in TI, k = nis_{li}, k' = nie_{mi}, (li, ls) = (mi, ms) \quad (3)$$

In addition, constraints (eqs 4–7) are developed for mass flows between substreams. Binary variables $ilis_{li,ls,k}$ and $olis_{li,ls,k}$ are used to represent the existence of importing and exporting mass flows of hot intermediate fluid substreams respectively, given by eqs 4 and 5

$$\sum_{ls' \in LS} flir_{li,ls',k} \leq MFIF \times ilis_{li,ls,k} \quad \forall (li, ls) \in HILS, k \in TI, nis_{li} < k \leq nie_{li} \quad (4)$$

$$\sum_{ls' \in LS} flir_{li,ls',k} \leq MFIF \times olis_{li,ls,k} \quad \forall (li, ls) \in HILS, k \in TI, nis_{li} < k \leq nie_{li} \quad (5)$$

where $MFIF$ represents the maximum flow rate of intermediate fluid streams. Note that importing and exporting mass flows of the same substream cannot coexist, given by eq 6. The flow rate of exporting mass flows must be smaller than the one of the corresponding substream, given by eq 7.

$$ilis_{li,ls,k} + olis_{li,ls,k} \leq 1 \quad \forall (li, ls) \in HILS, nis_{li} < k \leq nie_{li} \quad (6)$$

$$\sum_{ls' \in LS} flir_{li,ls',k} \leq flisk_{li,ls,k-1} \quad \forall (li, ls) \in HILS, k \in TI, nis_{li} < k \leq nie_{li} \quad (7)$$

Heat Flow Pattern of Intermediate Fluid Circles. In the heat flow pattern, each hot/cold intermediate fluid substream is considered as an individual stream. The heat flow pattern for a hot substream and a cold substream is shown in Figure 2-(b), which can be described as follows (take hot substreams as example): the residual energy $rqlis_{li,ls,k}$, which cannot be utilized in temperature interval $k - 1$, flows into temperature interval k ; the inherent energy $flisk_{li,ls,k}(tn_k - tn_{k+1})$ supplied by hot substream li, ls in temperature interval k ; the heat load $qlim_{li,ls,m,ms,k}$ of the heat transfer match flows out to a cold process substream m, ms ; the residual energy $rqlis_{li,ls,k+1}$, which cannot be utilized in temperature interval k , flows out to temperature interval $k + 1$. Energy balance formulation for each hot substream in each temperature interval is given as

$$flisk_{li,ls,k} \times (tn_k - tn_{k+1}) + rqlis_{li,ls,k} - rqlis_{li,ls,k+1} - \sum_{(m,ms) \in CPMS_p} qlim_{li,ls,m,ms,k} = 0 \quad \forall (li, ls) \in HILS, k \in TI, nis_{li} \leq k < nie_{li} \quad (8)$$

As mentioned before, intermediate fluid streams are transferred in a circle. The residual energy of hot substreams in the starting temperature level should be equal to the residual energy of the corresponding cold substreams in the ending temperature level

$$\begin{aligned}
 r q h i s_{l i, l s, k} &= 0 - r q c i s_{m i, m s, k} \\
 \forall (l i, l s) \in H I L S, (m i, m s) \in H I L S, \\
 (l i, l s) &= (m i, m s), k, k' \in T L, k = n i s_{l i}, k' = n i e_{m i}
 \end{aligned} \quad (9)$$

The Big-M formulations for the heat load of heat transfer matches between hot intermediate fluid substreams and cold process substreams are given by eq 10

$$\begin{aligned}
 q l i m_{l i, l s, m, m s, k} - \Omega l i m_{l i, m, k} \times z l i m_{l i, l s, m, m s, k} &\leq 0 \\
 \forall (l i, l s) \in H I L S, (m, m s) \in C P M S_p, k \in T I, \\
 n i s_{l i} \leq k < n i e_{l i}, k < n s_m
 \end{aligned} \quad (10)$$

where the binary variable $z l i m_{l i, l s, m, m s, k}$ represents the existence of heat transfer matches. In addition, the total heat load of heat transfer matches between hot intermediate substreams and cold process substreams must be equal to the one of heat transfers between cold intermediate substreams and hot process substreams, given by eq 11.

$$\begin{aligned}
 \sum_{(m, m s) \in C P M S_p} \sum_{l s \in L S} \sum_{k \in T I, n i s_{l i} \leq k < n i e_{l i}, k < n s_m} q l i m_{l i, l s, m, m s, k} \\
 = \sum_{(l i, l s) \in H P L S_p} \sum_{m s \in M S} \sum_{k \in T I, n i e_{m i} \leq k < n i s_{m i}, k \geq n s_l} q l i m_{l i, l s, m i, m s, k} \\
 \forall l i \in H I, m i \in C I, l i = m i
 \end{aligned} \quad (11)$$

In our previous transshipment model,³⁸ the residual energy is assumed to be non-negative variables. The residual energy of hot substreams represents the available energy in the specific temperature level, while the one of cold substreams represents the unsatisfied demand in the specific temperature level. The non-negativity of the residual energy is requisite for keeping every heat transfer match satisfying the minimum temperature approach. However, in this model, the residual energy of hot and cold intermediate fluid substreams can be both negative and positive. When the hot or cold intermediate fluid substream is not cooled down or heated up to its ending temperature, the residual energy will be negative on some temperature levels. An example is illustrated in Figure 3, where

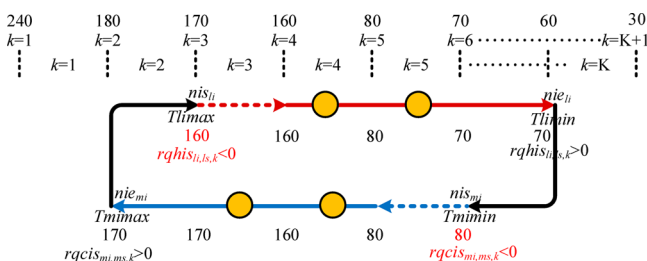


Figure 3. Illustration of an intermediate fluid circle.

the starting and ending temperatures of the hot intermediate substream are assumed to be 170 and 60 °C. Note that the temperatures of hot streams are subtracted by a $\Delta T_{\min}$ of 10 °C. It is assumed that the hot intermediate substream is only cooled down to 70 °C, resulting in a positive residual energy in the ending temperature level. When the residual energy of the hot intermediate fluid substream is positive in the ending temperature level $n i e$, the residual energy of the corresponding cold intermediate fluid substream will be negative, given by eq S9. That is, the starting temperature of the cold intermediate

substream (80 °C) is larger than $T m i m i n$ (70 °C). Similarly, since the ending temperature of the cold intermediate substream (170 °C) is smaller than $T m i m a x$ (180 °C), the starting temperature of the hot intermediate substream (160 °C) is smaller than $T l i m a x$ (170 °C), making the residual energy of the hot intermediate fluid substream negative given by eq 9. Under the circumstance, heat transfer matches are unavailable in the temperature intervals with negative residual energy. Heat transfer matches only exist when the residual energy of intermediate substreams in the corresponding temperature levels is non-negative. Thus, new formulations should be developed to identify heat transfer available temperature intervals. The binary variable $x l i s_{l i, l s, k}$ is used to indicate whether the residual energy of hot intermediate substreams is positive or negative

$$\begin{aligned}
 N M R L I_{l i, k} \times (1 - x l i s_{l i, l s, k}) &\leq r q h i s_{l i, l s, k} \leq P M R L I_{l i, k} \\
 &\times x l i s_{l i, l s, k} \\
 \forall (l i, l s) \in H I L S, k \in T I, n i s_{l i} \leq k < n i e_{l i}
 \end{aligned} \quad (12)$$

where the parameters $P M R L I_{l i, k}$ and $N M R L I_{l i, k}$ are the upper and lower bounds for the residual energy of hot intermediate fluid streams in the corresponding temperature level. When the residual energy is negative, $x l i s_{l i, l s, k}$ will be 0. On the contrary, $x l i s_{l i, l s, k}$ will be 1. When the residual energy is 0, $x l i s_{l i, l s, k}$ can be either 0 or 1. The binary variable $x l i s_{l i, l s, k}$ can be applied to identify whether hot intermediate substreams are available for heat transfer in the corresponding temperature interval. The heat transfer matches cannot exist when the residual energy is negative, as shown by the dotted line in Figure 3. Heat transfer matches only exist when the residual energy is non-negative

$$\begin{aligned}
 z l i m_{l i, l s, m, m s, k} &\leq x l i s_{l i, l s, k} \\
 \forall (l i, l s) \in H I L S, (m, m s) \in C P M S_p, k \in T I, \\
 n i s_{l i} \leq k < n i e_{l i}, k < n s_m \\
 z l i m_{l i, l s, m, m s, k} &\leq x l i s_{l i, l s, k+1} \\
 \forall (l i, l s) \in H I L S, (m, m s) \in C P M S_p, k \in T I, \\
 n i s_{l i} \leq k < n i e_{l i}, k < n s_m
 \end{aligned} \quad (13)$$

By eqs 12–14, heat transfer available temperature intervals can be identified, ensuring the availability of each heat transfer match. In addition, eq 15 is developed to keep the continuity of intermediate fluid streams.

$$x l i s_{l i, l s, k} \leq x l i s_{l i, l s, k+1} \quad \forall (l i, l s) \in H I L S, k \in T I, n i s_{l i} \leq k < n i e_{l i} \quad (15)$$

Additionally, it is known that the residual energy must be equal to 0 when there are exporting mass flows existing ($o l i s_{l i, l s, k} = 1$)

$$\begin{aligned}
 r q h i s_{l i, l s, k} &\leq P M R L I_{l i, k} \times (1 - o l i s_{l i, l s, k}) \quad \forall (l i, l s) \in H I L S, k \in T I, \\
 n i s_{l i} &< k \leq n i e_{l i}
 \end{aligned} \quad (16)$$

Meanwhile, when the residual energy is negative ($x l i s_{l i, l s, k} = 0$), there is no exporting mass flow existing in the corresponding temperature levels

$$x l i s_{l i, l s, k} \geq o l i s_{l i, l s, k} \quad \forall (l i, l s) \in H I L S, k \in T I, n i s_{l i} \leq k \leq n i e_{l i} \quad (17)$$

The negative residual energy will cause mathematical error in the objective function. Thus, the non-negative variable $p r q h i s_{l i, l s, k}$ is used to replace the residual energy $r q h i s_{l i, l s, k}$ in the

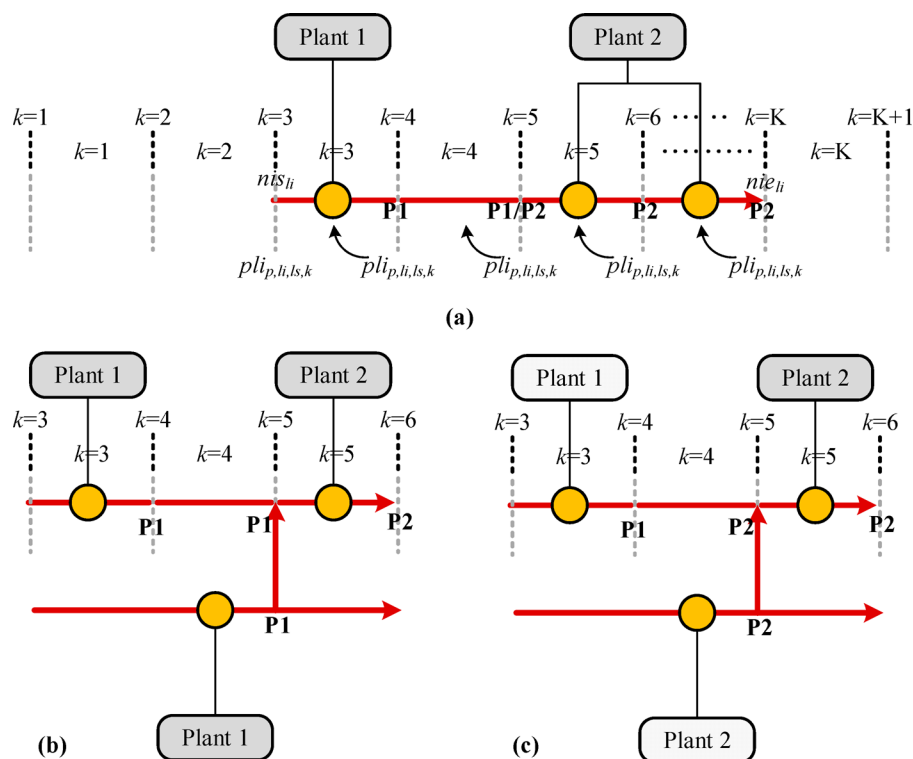


Figure 4. Illustration of location identification of intermediate fluid streams.

objective function. The variable $prqhis_{li,ls,k}$ will be equal to $rqhis_{li,ls,k}$ when $rqhis_{li,ls,k}$ is positive. On the contrary, the variable $prqhis_{li,ls,k}$ will be equal to zero, when $rqhis_{li,ls,k}$ is nonpositive. Accordingly, eqs 18–20 are formulated, where the residual energy equality is enforced if and only if $xlis_{li,ls,k}$ is equal to 1. The non-negative slack variable $srqhis_{li,ls,k}$ is (de)active by $xlis_{li,ls,k}$.

$$rqhis_{li,ls,k} + srqhis_{li,ls,k} = prqhis_{li,ls,k} \quad \forall (li, ls) \in \text{HILS}, k \in \text{TI},$$

$$nis_{li} \leq k \leq nie_{li} \quad (18)$$

$$srqhis_{li,ls,k} \leq -\text{NMRLI}_{li,k} \times (1 - xlis_{li,ls,k}) \quad \forall (li, ls) \in \text{HILS},$$

$$k \in \text{TI}, nis_{li} \leq k \leq nie_{li} \quad (19)$$

$$prqhis_{li,ls,k} \leq \text{PMRLI}_{li,k} \times xlis_{li,ls,k} \quad \forall (li, ls) \in \text{HILS}, k \in \text{TI},$$

$$nis_{li} \leq k \leq nie_{li} \quad (20)$$

Other constraints for heat transfer matches are given, such as only one heat transfer match for each hot intermediate substream in each temperature interval

$$\sum_{(m,ms) \in \text{CPMS}_p} zlim_{li,ls,m,ms,k} \leq 1 \quad \forall (li, ls) \in \text{HILS}, k \in \text{TI}, nis_{li} \leq k < nie_{li}$$

$$(21)$$

at least one heat transfer match for each hot intermediate fluid stream

$$\sum_{(m,ms) \in \text{CPMS}_p} \sum_{ls \in \text{LS}} \sum_{k \in \text{TI}, nis_{li} \leq k < nie_{li}, k < ns_m} zlim_{li,ls,m,ms,k} \geq 1 \quad \forall li \in \text{HI}$$

$$(22)$$

and only one heat transfer match for each pair of hot intermediate fluid substream and cold process substream.

$$\sum_{k \in \text{TI}, nis_{li} \leq k < nie_{li}, k < ns_m} zlim_{li,ls,m,ms,k} \leq 1 \quad \forall (li, ls) \in \text{HILS},$$

$$(m, ms) \in \text{CPMS}_p \quad (23)$$

Note that eq 23 can be replaced by a stricter constraint eq 23a

$$\sum_{ls \in \text{LS}} \sum_{ms \in \text{MS}} \sum_{k \in \text{TI}, nis_{li} \leq k < nie_{li}} zlim_{li,ls,m,ms,k} \leq 1 \quad \forall li \in \text{HI}, m \in \text{CP}_p$$

$$(23a)$$

which means that there is at most one heat exchanger for each pair of hot intermediate stream and cold process stream. The binary variable $zlim_{li,ls,m,ms,k}$ represents the existence of heat exchangers. The identification of heat exchangers is introduced in our previous work.³⁸ Also, the formulations can be found in the Supporting Information.

Interconnectivity Pattern of Intermediate Fluid Circles. In the proposed model, mass flows exist between hot/cold intermediate fluid substreams as shown in Figure 1. Meanwhile, each substream can exchange heat with process streams from different plants. An intermediate fluid circle may flow to several plants successively. The interconnectivity pattern of intermediate fluid circles is diverse. In this model, the interconnectivity pattern of intermediate fluid streams can be described according to the heat transfer matches. That is, heat transfer matches can determine the affiliation of hot/cold intermediate fluid substreams in all temperature levels. It is assumed that the affiliation of each substream in each temperature level is exclusive. Once the affiliation of each substream is known, the interconnectivity pattern of intermediate fluid circles can be easily obtained. The binary variable $pli_{p,li,ls,k}/pmi_{p,mi,ms,k}$ is used to denote whether the affiliation of hot/cold intermediate substream $li, ls/mi, ms$ in temperature level k is plant p .

It is obvious that the affiliation of hot intermediate substream li, ls in the starting temperature level is determined

by the one of cold intermediate substream mi , ms in the ending temperature level, given by eq 24. In the other temperature levels, it is determined by the heat transfer matches in their previous temperature interval, given by eqs 25–27.

$$p_{li,li,ls,k}^{li} = p_{mi,ms,k}^{mi} \quad \forall P \in PT, (li, ls) \in HILS, (mi, ms) \in CIMS, k, k' \in TI, k = nis_{li}, k' = nie_{mi} \quad (24)$$

$$p_{li,li,ls,k}^{li} \geq \sum_{(m,ms) \in CPMS_p} z_{lim_{li,ls,m,ms,k-1}} \quad \forall P \in PT, (li, ls) \in HILS, k \in TI, nis_{li} < k \leq nie_{li} \quad (25)$$

$$p_{li,li,ls,k}^{li} \leq \sum_{(m,ms) \in CPMS_p} z_{lim_{li,ls,m,ms,k-1}} + p_{li,ls,k-1}^{li} + p_{li,ls,k+1}^{li} \quad \forall P \in PT, (li, ls) \in HILS, k \in TI, nis_{li} < k \leq nie_{li} \quad (26)$$

$$\sum_{P \in PT} p_{li,li,ls,k}^{li} = 1 \quad \forall (li, ls) \in HILS, n \in TI, nis_{li} < k \leq nie_{li} \quad (27)$$

An example of a hot intermediate fluid substream is illustrated in Figure 4(a), while the data is listed in Table 1. The bold

Table 1. Corresponding Data of Figure 4(a)

k	3	4	5	6	K
$\sum_{(m,ms) \in CPMS_p} z_{lim_{li,ls,m,ms,k-1}} (p = 1)$	/	1	0	0	0
$\sum_{(m,ms) \in CPMS_p} z_{lim_{li,ls,m,ms,k-1}} (p = 2)$	/	0	0	1	1
$p_{li,li,ls,k}^{li} (p = 1)$	/	1	0/1	0	0
$p_{li,li,ls,k}^{li} (p = 2)$	/	0	1/0	1	1
$\sum_{P \in PT} p_{li,li,ls,k}^{li}$	/	1	1	1	1

labels P1 and P2 in Figure 4 represent plant 1 and plant 2, respectively. As shown in Figure 4(a), the hot intermediate substream li , ls exchanges heat with a cold process substream of plant 1 in temperature interval 3; therefore, the binary variable $p_{li,li,ls,k=4}^{li}$ will be equal to 1, given by eq 25. Similarly, the binary variable $p_{li,li,ls,k=6}^{li}$ and $p_{li,li,ls,k=K}^{li}$ will be equal to 1, according to the heat transfer matches with process streams of plant 2. As for temperature interval 4, in which there is no heat transfer match, the binary variable $p_{li,li,ls,k=5}^{li}$ will be equal to the one of the previous temperature level $p_{li,li,ls,k=4}^{li}$ or the one of the next temperature level $p_{li,li,ls,k=6}^{li}$, given by eq 26. It will be optimized automatically to reduce the piping cost and pumping cost. Two cases are illustrated in Figure 4(b,c). As for the case in Figure 4(b), when the affiliation of temperature level 5 is P1, there only needs to be one pump to transport the intermediate fluid substream from plant 1 to plant 2. Otherwise, there will be two pumps needed. As for the case in Figure 4(c), the affiliation of temperature level 5 must be P2, otherwise, unnecessary piping cost and pumping cost will be incurred. Finally, the affiliation in each temperature level should be exclusive, given by eq 27.

With the affiliation of hot intermediate substream li , ls in each temperature level, whether pipes and pumps are needed in each temperature interval can be determined by eqs 28–31.

$$flisk_{li,ls,k} \leq mfif \times zflisk_{li,ls,k} \quad \forall (li, ls) \in HILS, nis_{li} \leq k < nie_{li} \quad (28)$$

$$ppli_{li,ls,k} \geq pli_{p,li,ls,k} + pli_{p',li,ls,k+1} + zflisk_{li,ls,k} - 2 \quad \forall p, p' \in PT, (li, ls) \in HILS, k \in TI, nis_{li} \leq k < nie_{li}, p \neq p' \quad (29)$$

$$ppli_{li,ls,k} \leq zflisk_{li,ls,k} \quad \forall (li, ls) \in HILS, k \in TI, nis_{li} \leq k < nie_{li} \quad (30)$$

$$ppli_{li,ls,k} \leq 2 - pli_{p,li,ls,k} - pli_{p',li,ls,k+1} \quad \forall p, p' \in PT, (li, ls) \in HILS, k \in TI, nis_{li} \leq k < nie_{li}, p = p' \quad (31)$$

The binary variable $ppli_{li,ls,k}$ is used to denote the existence of pipe and pump for hot intermediate substream li , ls in temperature interval k , while the binary variable $zflisk_{li,ls,k}$ represents the existence of hot intermediate substream li , ls in temperature interval k . The binary variable $ppli_{li,ls,k}$ will be equal to 1, only and if only the affiliation of two adjacent temperature levels is different and the substream is existing in the corresponding temperature interval, given by eq 29. When the substream is not existing in the corresponding temperature interval or the affiliation of two adjacent temperature levels are the same, $ppli_{li,ls,k}$ will be equal to 0, given by eqs 30 and 31, respectively. Also, with eqs 32–34, the transporting distance $ddli_{li,ls,k}$ in each temperature interval can be determined. In addition, the existence of pipe and pump for mass flows between hot intermediate substreams $ppflir_{li,ls,ls',k}$ and the transporting distance $ddflir_{li,ls,ls',k}$ can be determined by eqs 35–43.

$$ddli_{li,ls,k} \geq (pli_{p,li,ls,k} + pli_{p',li,ls,k+1} + zflisk_{li,ls,k} - 2) \times dd_{p,p'} \quad \forall p, p' \in PT, (li, ls) \in HILS, k \in TI, nis_{li} \leq k < nie_{li}, p \neq p' \quad (32)$$

$$ddli_{li,ls,k} \leq zflisk_{li,ls,k} \times Mdd \quad \forall (li, ls) \in HILS, k \in TI, nis_{li} \leq k < nie_{li} \quad (33)$$

$$ddli_{li,ls,k} \leq (2 - pli_{p,li,ls,k} - pli_{p',li,ls,k+1}) \times Mdd + dd_{p,p'} \quad \forall p, p' \in PT, (li, ls) \in HILS, k \in TI, nis_{li} \leq k < nie_{li} \quad (34)$$

$$flir_{li,ls,ls',k} \leq MFIF \times zflir_{li,ls,ls',k} \quad \forall (li, ls), (li, ls') \in HILS, k \in TI, nis_{li} < k \leq nie_{li} \quad (35)$$

$$zflir_{li,ls,ls',k} \leq ilis_{li,ls',k} \quad \forall (li, ls), (li, ls') \in HILS, k \in TI, nis_{li} < k \leq nie_{li} \quad (36)$$

$$zflir_{li,ls,ls',k} \leq olis_{li,ls,k} \quad \forall (li, ls), (li, ls') \in HILS, k \in TI, nis_{li} < k \leq nie_{li} \quad (37)$$

$$ppflir_{li,ls,ls',k} \geq pli_{p,li,ls,k} + pli_{p',li,ls',k} + zflir_{li,ls,ls',k} - 2 \quad \forall (li, ls), (li, ls') \in HILS, k \in TI, nis_{li} < k \leq nie_{li}, p \neq p' \quad (38)$$

$$ppflir_{li,ls,ls',k} \leq zflir_{li,ls,ls',k} \quad \forall (li, ls), (li, ls') \in HILS, k \in TI, nis_{li} < k \leq nie_{li} \quad (39)$$

$$ppflir_{li,ls,ls',k} \leq 2 - pli_{p,li,ls,k} + pli_{p',li,ls',k} \quad \forall p, p' \in PT, (li, ls), (li, ls') \in HILS, k \in TI, nis_{li} < k \leq nie_{li}, p = p' \quad (40)$$

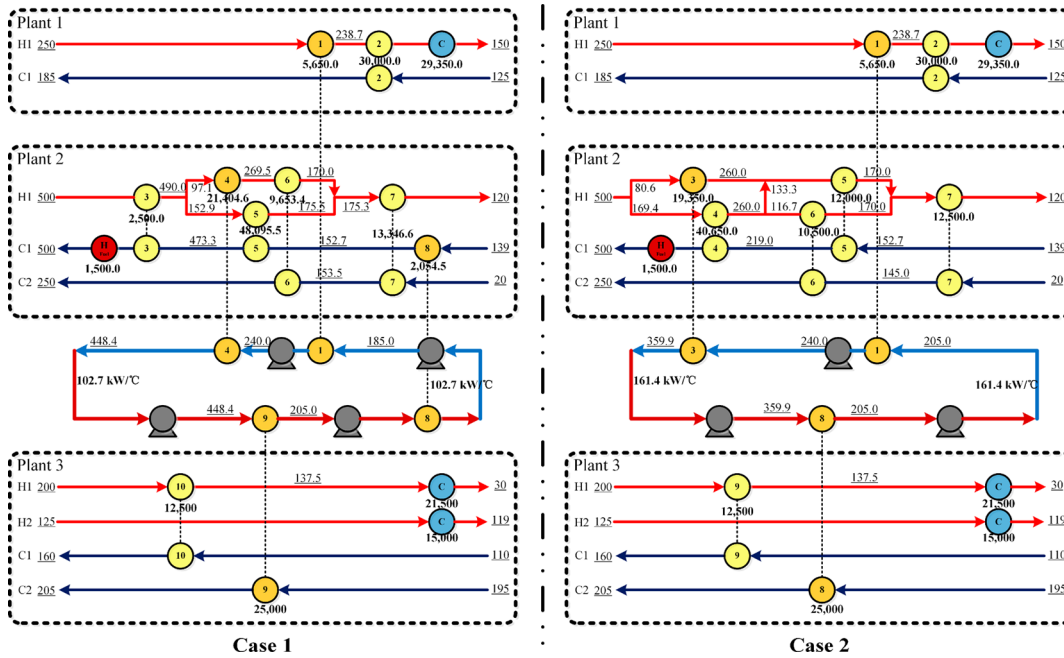


Figure 5. Intra and interplant HEN for example 1 (cases 1 and 2).

$$ddflir_{li,ls,ls',k} \geq (pli_{p,li,ls,k} + pli_{p',li,ls',k} + zflir_{li,ls,ls',k} - 2) \times dd_{p,p'} \\ \forall p, p' \in PT, (li, ls), (li, ls') \in HILS, k \in TI, nis_{li} < k \leq nie_{li}, p \neq p' \quad (41)$$

$$ddflir_{li,ls,ls',k} \leq zflir_{li,ls,ls',k} \times Mdd \quad \forall (li, ls), (li, ls') \in HILS, \\ k \in TI, nis_{li} < k \leq nie_{li} \quad (42)$$

$$ddflir_{li,ls,ls',k} \leq (2 - pli_{p,li,ls,k} + pli_{p',li,ls',k}) \times Mdd + dd_{p,p'} \\ \forall p, p' \in PT, (li, ls), (li, ls') \in HILS, k \in TI, nis_{li} < k \leq nie_{li}, p = p' \quad (43)$$

It is known that hot/cold intermediate fluid streams can exchange heat with cold/hot process streams in any plants among the heat transfer available temperature intervals. However, high degree of freedom will make the interconnectivity pattern complex and impractical and also increase the computational burden. Thus, several constraints are developed to simplify the interconnectivity pattern. It is assumed that each hot intermediate fluid substream can only be transported between different plants twice

$$\sum_{k \in TI, nis_{li} \leq k < nie_{li}} ppli_{li,ls,k} \leq \min(2, P - 1) \quad \forall (li, ls) \in HILS \quad (44)$$

where P represents the number of plants. Similarly, the mass flow between substreams can only be transported between different plants twice

$$\sum_{\forall (li,ls),(li,ls') \in HILS} \sum_{k \in TI, nis_{li} < k \leq nie_{li}} ppflir_{li,ls,ls',k} \leq \min(2, P - 1) \quad \forall li \in HI \quad (45)$$

Then, eqs 46 and 47 are used to ensure that only one hot intermediate fluid stream is involved in each plant

$$zpli_{p,li} \geq zlim_{p,li,ls,ms,k} \quad \forall p \in PT, li \in HI, (li, ls) \in HILS, \\ (p, m) \in CP, (p, m, ms) \in CMS, k \in TI, nis_{li} \leq k < nie_{li}, k < ns_{p,m} \quad (46)$$

$$zpli_{p,li} + zpli_{p,li'} \leq 1 \quad \forall p \in PT, (li, ls), (li, ls') \in HILS \quad (47)$$

where the integer variable $zpli_{p,li}$ represents whether hot intermediate fluid stream li exchanges heat with any cold process streams in plant p . Additionally, eqs 48 and 49 are developed to limit the maximum number of involved plants for each intermediate fluid circle. In this research, we assume that each intermediate fluid circle can only be shared by at most three plants. Similarly, a set of constraints for cold intermediate fluid streams are given by eqs S1–S49 in the Supporting Information.

$$zplimi_{p,li,mi} \geq zpli_{p,li} \quad \forall p \in PT, li \in HI, mi \in CI, li = mi \quad (48)$$

$$\sum_{p \in PT} zplimi_{p,li,mi} \leq \min(3, P) \quad \forall li \in HI, mi \in CI, li = mi \quad (49)$$

Objective Function. The objective function, the total annual cost, is shown in eq 50, which includes the hot utility cost (HUC), the cold utility cost (CUC), the heat exchanger cost (EXC), the total piping cost (PIC), and the total pumping cost (PMC). The piping and pumping cost are calculated by the method from Chang et al.³⁰ The formulations for the HUC, CUC, EXC, PIC, and PMC are given by eqs S51–S53, S69, and S70, respectively, in the Supporting Information.

$$\min TAC = CUC + HUC + EXC + PIC + PMC \quad (50)$$

CASE STUDIES AND DISCUSSIONS

In this section, three examples with different complexities are illustrated to show the applicability of the proposed model. The data of process streams and utilities of three examples are given in Tables S1–S3, respectively. The parameters of heat exchanger, pump, intermediate fluid, the annual factor, and the electric price are listed in Table S4, adopted from Chang et al.³⁰ The proposed model is carried out in GAMS 24.9



we adopt GUROBI and CPLEX. The model data of three examples are provided in [Table S5](#).

Example 1: The Effect of Interconnectivity Patterns. The first example is a three-plant problem, adopted from Bader and Bandyopadhyay.²⁷ The distances among three plants are

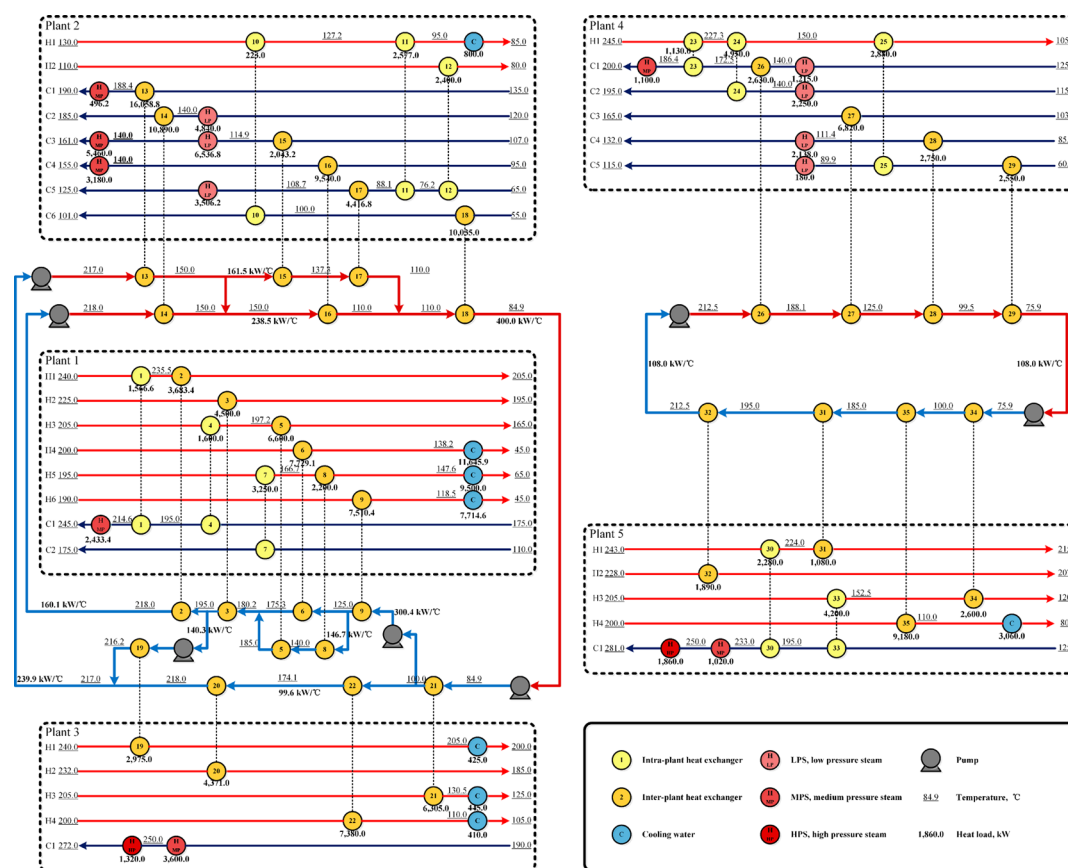


Figure 7. Intra and interplant HEN for Example 3.

$L_{12} = 1.0$ km, $L_{13} = 2.0$ km, and $L_{23} = 1.0$ km. The minimum temperature difference $\Delta T_{\min}$ for both intra and interplant heat integration is 10.0 °C. There are two, four, and four process streams in the three plants, respectively. Since the number of process streams is small, the capital cost of the pipeline incurred by the implementation of interplant can be a major part of the total cost. Also, each plant can be both a heat sink and heat source simultaneously. The interconnectivity patterns will be of significance to the performance of heat integration. In this example, four cases are illustrated in the following to demonstrate the effect of interconnectivity patterns.

First of all, the temperature intervals of the transshipment model must be constructed. According to the starting and ending temperatures of all process streams, it is known that $T_{\max}$, $T_{\min}$, $T_{\max}$, and $T_{\min}$ are 480, 20, 490, and 30 °C, respectively. Then, the whole temperature intervals can be constructed: 990–490–480–290–250–240–205–195–190–185–160–139–125–115–110–109–30–20–15.6 (°C), with both the starting and ending temperatures of process streams. Next, the model is tested on two cases: case 1, where each plant can exchange heat with both hot and cold intermediate fluid streams, and case 2, where each plant can only exchange heat with either hot or cold intermediate fluid streams. The HENs of cases 1 and 2 are shown in Figure 5. This example is also studied by Chang et al.,³⁰ while the result comparisons are shown in Table S6. It indicates that the results of all cases are better than the one of Chang et al.,³⁰ while the TAC of the best one (case 1) decreases 4.4%. In case 1, when every plant is allowed to exchange heat with both hot and cold intermediate fluid streams, plant 2 can be both the heat sink and heat source. The intermediate fluid stream extracts heat from plants

1 and 2 and then release heat to plants 3 and 2 in case 1, while the intermediate fluid stream transfers heat from plants 1 and 2 to plant 3 in case 2. Although one more pump is needed in case 1, the flow rate of the intermediate fluid stream is much smaller, compared to case 2. In addition, the piping cost is linear to the distance. The distance between plants 1 and 3 is twice of the distance between plants 1 and 2 and plants 2 and 3. In case 2, the intermediate fluid stream is transported from plant 1 to plant 3. In case 1, the intermediate fluid stream is transported from plant 3 to plant 2 and then to plant 1. Even when the flow rate of the intermediate fluid stream of cases 1 and 2 are the same, the piping cost of case 1 will not be larger than case 2. Although the interconnectivity pattern of case 1 is more complex than the one of case 2, the total annual cost is still smaller owing to the smaller piping cost.

In addition, we also have Cases 3 and 4 in another situation with less temperature intervals. The temperature intervals are constructed without the ending temperature of all streams, which will be 990–490–480–290–240–195–190–139–125–115–110–30–20–15.6 (°C). The results are also provided in Table S6. It indicates that the utility consumption of case 3 is smaller than the one of case 4. Similar to the comparisons between cases 1 and 2, the total annual cost of case 3 is smaller than the one of case 4, even though one more pump is needed in case 3. It can be seen that the interconnectivity pattern is one of the significant impacts when considering the performance of interplant heat integration. However, more interconnectivity patterns not only bring more heat integration possibilities but also increase the number of variables (Table S5). This problem will be more obvious when the number of temperature intervals is large,

especially for large-scale problems. Additionally, from the comparisons between cases 1 and 3 and cases 2 and 4, it is known that the results obtained by cases 1 and 2 with more temperature intervals are better than the ones of cases 3 and 4 with less temperature intervals. However, more temperature intervals result in more variables. It is suggested to use only starting temperatures to construct temperature intervals for large-scale problems. Additionally, for small-scale problems, using both starting and ending temperatures would be a better option. The construction of temperature intervals could also be found in our previous research.³⁸

Example 2: The Effect of More Structures. The second example was adopted from Rodera and Bagajewicz⁴⁰ and includes two plants. The distance between two plants is 1.0 km. The minimum temperature difference $\Delta T_{\min}$ for both intra and interplant heat integration is 5.6 °C. Then, according to the starting and ending temperatures of all process streams, $T_{\max}$, $T_{\min}$, $T_{\max}$, and $T_{\min}$ for intermediate fluid circles can be determined, which are 337, 30, 342.6, and 35.6 °C, respectively. In this example, only the ending temperature of the process streams is adopted to construct temperature intervals, resulting in 30 temperature intervals. By solving the proposed model, a HEN with an intermediate fluid circle of 172.6 kW/°C can be obtained, as shown in Figure 6.

This example was also studied in the research of Rodera and Bagajewicz⁴⁰ and Chang et al.³⁰ In the research of Rodera and Bagajewicz,⁴⁰ they only provide the utility consumption target and the location of intermediate fluid circles. The result comparisons are shown in Table S7. It can be seen that energy savings achieved by a single circle of Chang et al.³⁰ and the proposed model are larger than the one obtained by Rodera and Bagajewicz.⁴⁰ It is because Rodera and Bagajewicz⁴⁰ did not adopt the total annual cost as an objective function. In addition, although the flow rate of the intermediate fluid circle and the utility consumption of the proposed model are larger than the ones of Chang et al.,³⁰ the TAC is still slightly better than the one of Chang et al.³⁰ The reason is that several structures in Figure 6, such as non-isothermal mixing and bypass, are not included in the model of Chang et al.³⁰ The heat exchanger cost reduction leads to the smaller TAC.

Example 3: The Effect of Multiple Utilities. The third example is adopted from Chang et al.,³⁰ which is an industrial scale heat integration project. Five plants, an aromatic plant, a nitrobenzene plant, a hydrogen plant, a delayed coking plant, and an aniline plant, are included in the example. The distances between each pair of plant are $L_{12} = 1500$ m, $L_{13} = 300$ m, $L_{14} = 1300$ m, $L_{15} = 1900$ m, $L_{23} = 1800$ m, $L_{24} = 600$ m, $L_{25} = 400$ m, $L_{34} = 1000$ m, $L_{35} = 1200$ m, and $L_{45} = 200$ m. The minimum temperature difference $\Delta T_{\min}$ for both intra and interplant heat integration is 10.0 °C. There can be many alternative interconnectivity patterns among the five plants. This example is illustrated to demonstrate that the proposed model can be applied to an industrial scale problem, and a promising interconnectivity pattern with a good performance of heat integration can be obtained. Also, the advantage of multiple utilities will be highlighted in this example.

The obtained result with the TAC of 4 824 345 \$/y is shown in Figure 7. The total heat capacity flow rate of the intermediate fluid stream going through plants 1, 2, and 3, named IF1, is 400 kW/°C. The intermediate fluid stream IF1 extracts heat from plant 1 and plant 3 and releases heat to plant 2. For the intermediate fluid stream between plants 4 and 5, named IF2, the heat capacity flow rate is 108 kW/°C, while

plants 4 and 5 are the heat sink and heat source, respectively. The total hot utility consumption and cold utility consumption are 41 135 and 34 001 kW, respectively. From the result comparisons in Table S8, it can be seen that the hot utility cost of Figure 7 is 14% smaller than the one of Chang et al.,³⁰ although the total consumption is 18% larger. The reason is that multiple utilities are available for process streams in the proposed model. However, only one utility is available in the outlet of each process stream in Chang et al.,³⁰ where the stage-wise superstructure is adopted. As shown in Figure 7, there are three cold process streams consuming two kinds of hot utilities, and three cold process streams are preheated by low pressure steam before exchanging heat with other hot streams. As shown in Table 2, cheaper low pressure steam is

Table 2. Utility Consumption Comparisons between Literature and This Research

	Chang et al. ³⁰	this research
LPS, kW (40 \$/kW-y)	3653	20 666
MPS, kW (100 \$/kW-y)	25 851	17 289
HPS, kW (150 \$/kW-y)	5400	3180
fuel, kW (200 \$/kW-y)	0	0
total, kW	34 904	41 135
HUC, \$/y	3 541 192	3 032 576

consumed, resulting in less medium pressure steam and high pressure steam consumption, which are much more expensive than low pressure steam. Thus, the total hot utility cost is much lower than the one of the literature.³⁰ It indicates that the implementation of multiple utilities has a significant impact in reducing utility cost. In addition, two intermediate fluid circles are included in both Figure 7 and the result of Chang et al.,³⁰ but the interconnectivity patterns are different. In Figure 7, plant 3 is integrated with plants 1 and 2, while plant 3 is integrated with plants 4 and 5 in Chang et al.³⁰ The flow rate of IF1 is only slightly larger than the one of Chang et al.,³⁰ when plant 3 is integrated with plants 1 and 2. However, the flow rate of IF2 is much smaller than the one of Chang et al.,³⁰ when plant 3 is not integrated with plants 4 and 5. Thus, the piping cost caused by transporting intermediate fluid streams is less than the one of Chang et al.³⁰ Additionally, the heat exchanger cost of Figure 7 reduces 3.9%, compared to the literature.³⁰ The result comparisons show that the trade-offs between the utility consumption (cost) and the piping and pumping cost are well-optimized in the proposed model, which leads to better interconnectivity patterns.

CONCLUSION

The energy efficiency can be further improved by applying interplant heat integration among industrial clusters. Simultaneous synthesis of intra and interplant HEN would be of significance to energy saving and emission reduction of process industries. In this paper, a new transshipment type HEN model for both intra and interplant heat integration using intermediate fluid circles is proposed. The utility cost, the heat exchanger cost, the piping cost, and the pumping cost are simultaneously optimized in the model. New mixed-integer linear formulations based on the mass flow pattern and heat flow pattern of transshipment model are developed to describe intermediate fluid circles, whose temperatures and flow rate are both unknown. Temperature interval distribution of intermediate fluid circles is determined according to the temper-

atures of all process streams, while heat transfer available temperature intervals can be identified by new formulations. By the proposed model, each intermediate fluid circle may exchange heat with any process stream from any plant, which means that interconnectivity patterns of intermediate fluid circles can be widely explored. In addition, newly generated mixed-integer linear formulations are used to follow up the interconnectivity patterns so as to optimize piping and pumping cost. Not only the energy efficiency of different heat integration schemes but also the total annual cost is considered and optimized in the proposed model.

The model is formulated as an MINLP problem with all constraints being linear. Three cases are illustrated to show the effectiveness of the proposed transshipment type model for both intra and interplant heat integration. The results indicate that better results with lower TAC can be obtained, compared to the literature results. The trade-offs among the utility cost, the heat exchanger cost, and the piping and pumping cost are well-optimized. For the industrial scale problems with five plants, appropriate interconnectivity patterns of intermediate fluid circles can be determined, with good performance of heat integration.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.9b00440.

Tables of stream data and parameters; tables of result comparisons between literature and this research; formulations for mass flow pattern of intermediate fluid circles, heat flow pattern of intermediate fluid circles, interconnectivity pattern of intermediate fluid circles, objective function, mass flow pattern of intraplant HEN, heat flow pattern of intraplant HEN, identification of heat exchangers, temperature differences for heat exchangers, bounds and fixed value for variables (PDF)

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Notes

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■ NOTATION

Subscripts

k, k' temperature interval, temperature level
 l, l' hot stream
 li, li' hot intermediate stream

ls, ls' hot stream substream
 m, m' cold stream
 mi, mi' cold intermediate stream
 ms, ms' cold stream substream
 p, p' plant

Sets

C_p cold stream m in plant p including process and utility streams, $C_p = CP_p \cup CU_p$, $C_p = (mlm = 1, 2, 3 \dots M)$, M represents utility with lowest temperature
 CP_p hot process stream m in plant p , $CP_p \subset C_p$
 CU_p cold utility m in plant p , $CU_p \subset C_p$
 CMS_p cold substream m, ms in plant p , $CMS_p = ((m, ms) | (m) \in C_p, ms \in MS)$, $CMS_p = CPMS_p \cup CUMS_p$
 $CPMS_p$ cold substream m, ms in plant p , $CPMS_p = ((m, ms) | m \in CP_p, ms \in MS)$
 $CUMS_p$ cold substream m, ms in plant p , $CUMS_p = ((m, ms) | m \in CU_p, ms \in MS)$
 CI cold intermediate stream mi
 $CIMS$ cold intermediate substream mi, ms , $CIMS = ((mi, ms) | mi \in CI, ms \in MS)$
 H_p hot stream l in plant p including process and utility streams, $H_p = HP_p \cup HU_p$, $H_p = (lll = 1, 2, 3 \dots L)$, L represents the utility with highest temperature
 HP_p hot process stream l in plant p , $HP_p \subset H_p$
 HU_p hot utility l in plant p , $HU_p \subset H_p$
 HLS_p hot substream l, ls in plant p , $HLS_p = ((l, ls) | l \in H_p, ls \in LS)$, $HLS_p = HPLS_p \cup HULS_p$
 $HPLS_p$ hot substream l, ls in plant p , $HPLS_p = ((l, ls) | l \in HP_p, ls \in LS)$
 $HULS_p$ hot substream l, ls in plant p , $HULS_p = ((l, ls) | l \in HU_p, ls \in LS)$
 HI hot intermediate stream li
 $HILS$ hot intermediate substream li, ls , $HILS = ((li, ls) | li \in HI, ls \in LS)$
 MS substream of cold stream, $MS = (mslms = 1, 2, 3 \dots)$
 LS substream of hot stream, $LS = (lslls = 1, 2, 3 \dots)$
 TI temperature interval, $TI = (klk = 1, 2, 3 \dots K)$, resulting $K + 1$ temperature levels
 P plant, $PT = (plp = 1, 2, 3 \dots P)$

Continuous Variables

Mass Flow Pattern

$flif_{li}$ flow rate of hot/cold intermediate stream li/mi
 $flir_{li,ls,ls',k}$ flow rate of mass flow from substreams li, ls to li, ls' in temperature level k
 $flisk_{li,ls,k}$ flow rate of substream $li, ls/mi, ms$ in temperature interval k

Heat Flow Pattern

$qlim_{li,ls,m,ms,k}$ heat load of heat exchanger between substreams li, ls and m, ms in temperature interval k
 $qlmi_{li,ls,mi,ms,k}$ heat load of heat exchanger between substreams l, ls and mi, ms in temperature interval k
 $rqhis_{li,ls,k}$ residual energy of substream li, ls in temperature level k
 $rqcis_{mi,ms,k}$ residual energy of substream mi, ms in temperature level k
 $srqhis_{li,ls,k}$ slack residual energy of substream $li, ls/mi, ms$ in temperature level k
 $prqhis_{li,ls,k}$ positive residual energy of substream $li, ls/mi, ms$ in temperature level k

Interconnectivity Pattern of Intermediate Fluid Circles

$dli_{li,ls,k}$ pumping distance for substream $li, ls/mi, ms$ in temperature interval k

$dflir_{li,ls',k}$ pumping distance for mass flow between substreams in temperature level k

Objective Function

CUC/HUC cold/hot utility cost
 EXC heat exchanger cost
 PIM piping cost
 PMC pumping cost
 TAC total annual cost

Discrete Variables

$ilisk_{li,ls,k}$ existence of importing flow for substream li , ls in temperature level k
 $olisk_{li,ls,k}$ existence of exporting flow for substream li , ls in temperature level k
 $pli_{p,li,ls,k}$ index for affiliation of substream li , ls in temperature level k
 $pmi_{p,mi,ms,k}$ index for affiliation of substream mi , ms in temperature level k
 $ppli_{li,ls,k}$ existence of pump for substream li , ls in temperature interval k
 $ppflir_{li,ls',k}$ existence of pump for mass between substreams in temperature level k
 $xlisk_{li,ls,k}$ index of residual energy for substream li , ls in temperature level k
 $zlim_{li,ls,m,ms,k}$ existence of heat transfer match between substreams li , ls and m , ms in temperature interval k
 $zflir_{li,ls',k}$ existence of mass flow between substreams li , ls and li , ls' in temperature level k
 $zflisk_{li,ls,k}$ existence of substream li , ls in temperature interval k
 $zzlim_{li,ls,m,ms,k}$ existence of heat exchanger between substreams li , ls and m , ms in temperature interval k
 $zpli_{p,li}$ existence of heat transfer matches between hot intermediate fluid stream li and any cold process stream in plant p
 $zplimi_{p,li,mi}$ existence of heat transfer matches between hot/cold intermediate fluid stream li/mi and any cold/hot process stream in plant p

Parameters

$d_{p,p'}$ distance between plant p and p'
 $MFIF$ maximum flow rate for intermediate stream
 ne_l/ns_l ending/starting temperature level of hot stream l , ne_p , $ns_l \in TL$
 ne_m/ns_m ending/starting temperature level of cold stream m , ne_m , $ns_m \in TL$
 nie_{li}/nis_{li} ending/starting temperature level of hot intermediate fluid li , nie_{li} , $nis_{li} \in TL$
 nie_{mi}/nis_{mi} ending/starting temperature level of cold intermediate fluid mi , nie_{mi} , $nis_{mi} \in TL$
 $PMRLI_{li,k}$ maximum residual energy of hot intermediate stream li in temperature level k
 $NMRLI_{li,k}$ minimum residual energy of hot intermediate stream li in temperature level k
 tn_k temperature of temperature level k
 Tl_{max}/Tl_{min} maximum/minimum temperature for hot intermediate fluid
 Tm_{max}/Tm_{min} maximum/minimum temperature for cold intermediate fluid
 $\Omega_{lim_{li,m,k}}$ maximum heat load of heat transfer matches between hot intermediate stream li and cold process stream m in temperature interval k

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